KEMENTERIAN PENDIDIKAN TINGGI
JABATAN PENDIDIKAN POLITEKNIK DAN KOLEJ KOMUNITI

JABATAN TEKNOLOGI MAKLUMAT DAN KOMUNIKASI

DIPLOMA TEKNOLOGI MAKLUMAT
(TEKNOLOGI DIGITAL)

FINAL YEAR PROJECT
PROPOSAL REPORT

Omni-View Business Command Centre

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1.0 INTRODUCTION

Capt Empire is a retail business specializing in the sale of shirts through both physical and online sales channels. As the business grows, managing daily operations such as sales records, inventory, employee performance, and Key Performance Indicators (KPIs) becomes increasingly important. Relying on manual record-keeping or multiple disconnected tools can lead to inaccurate data, inefficient workflows, and difficulties in monitoring overall business performance.

To address these challenges, this project proposes the development of a web-based Integrated Operations Management System for Capt Empire. The system is designed to provide a centralized platform for managing the company's internal operations by integrating multiple business functions into a single application. This enables management to efficiently monitor sales performance, inventory levels, employee productivity, and operational KPIs.

The proposed system will include modules for sales management, inventory management, employee management, KPI tracking, and reporting. An administrative dashboard will present key business information through charts and summaries, allowing management to analyze sales trends, evaluate employee performance, monitor stock levels, and make informed business decisions.

By digitalizing and integrating operational processes, the proposed system is expected to improve data accuracy, reduce manual workload, enhance operational efficiency, and support data-driven decision-making. Ultimately, the system aims to strengthen Capt Empire's business management by providing a reliable and user-friendly platform for its daily operations.

2.0 PROBLEM STATEMENT

Capt Empire currently relies on manual processes and separate records to manage its daily business operations. Sales transactions, inventory updates, and employee performance data are not maintained within a centralized system, making it difficult to access accurate and up-to-date information. This fragmented approach increases the risk of data inconsistencies, duplicate records, and human errors.

Furthermore, the company lacks an effective mechanism to monitor employee performance and Key Performance Indicators (KPIs). Without a centralized dashboard, management faces difficulties in evaluating staff productivity, measuring business performance, and identifying areas that require improvement. Preparing reports and analyzing sales trends also require significant manual effort, reducing operational efficiency and delaying decision-making.

In addition, inventory monitoring is not integrated with sales records, making it challenging to maintain accurate stock information and identify fast-moving or low-stock products. This limits the company's ability to plan inventory effectively and optimize business operations.

Therefore, there is a need for a web-based Integrated Operations Management System that centralizes sales management, inventory control, employee performance monitoring, KPI tracking, and business reporting. Such a system will improve operational efficiency, enhance data accuracy, and provide management with valuable insights for informed decision-making.

3.0 OBJECTIVE

1. **Develop a web-based Integrated Operations Management System:**

   System for Capt Empire that centralizes sales, inventory, employee, and KPI management.

2. **Implement sales and inventory management modules:**

   Enable efficient recording, monitoring, and analysis of business transactions and stock levels.

3. **Develop an employee performance and KPI monitoring module:**

   Assists management in evaluating staff productivity and operational performance.

4. **Provide an administrative dashboard :**

   Provide a dashboard with reports and data visualizations that support business analysis and informed decision-making.

4.0 SCOPE

Admin Dashboard Features:

1. Product management (Create, Read, Update, Delete operations)
2. Stock inventory management with easy-to-use interface for status updates
3. Sales records tracking (transaction date, product, quantity, revenue)
4. Staff performance dashboard displaying basic metrics (sales volume, transaction count)
5. Admin user authentication and access control
6. Staff are able to interact with an AI-powered chatbot to obtain answers to sales-related questions.
7. Database backend with relational data structure

Technical Implementation:

1. Full-stack web application (frontend + backend)
2. Relational database for data persistence
3. Server-side business logic and data validation

Customer-Facing Features:

1. Product catalog with display of item details (name, description, images, price)
2. Near-real-time stock status display (Available, Limited Stock, Sold Out)
3. Size reference chart and/or sizing calculator tool
4. Direct redirect links to Shopee and TikTok Shop for customer checkout
5. Responsive web interface for browsing

5.0 PROJECT SIGNIFICANT

This system directly addresses operational bottlenecks that impact customer satisfaction and business efficiency. By centralizing stock visibility and reducing customer inquiries through live chat, the company can allocate support resources more effectively. The integrated analytics dashboard provides actionable business intelligence for informed decision-making, enabling management to track sales trends and staff performance in real time.

Customer Experience Improvement:

Customers benefit from immediate, self-service access to product availability and accurate sizing information, reducing friction in the shopping journey and decreasing return rates due to sizing errors. This improves customer satisfaction and builds trust in the brand.

Operational Efficiency:

Staff can manage inventory updates seamlessly through an intuitive admin interface. Management gains visibility into staff performance metrics, enabling data-driven performance evaluation and resource allocation.

Technical Relevance:

The project demonstrates full-stack web development proficiency, integrating frontend technologies (HTML, CSS, JavaScript, Bootstrap) with modern backend infrastructure (Supabase/PostgreSQL). It showcases practical experience in database design, authentication systems, and real-time data management — skills critical in contemporary e-commerce development.

Scalability & Future Growth:

The modular architecture allows for future enhancements, such as advanced analytics, inventory forecasting, or API integrations with e-commerce platforms, positioning the company for sustainable growth.

6.0 LITERATURE REVIEW

Modern retail businesses increasingly rely on integrated digital systems to manage customer interactions and operational efficiency. E-commerce platforms serve dual purposes: providing customers with self-service access to product information and inventory, while simultaneously generating data that informs business decisions (Chaffey & Ellis-Chadwick, 2019). Research indicates that real-time inventory visibility significantly reduces customer support costs and improves customer satisfaction (Kim & Christodoulidou, 2020).

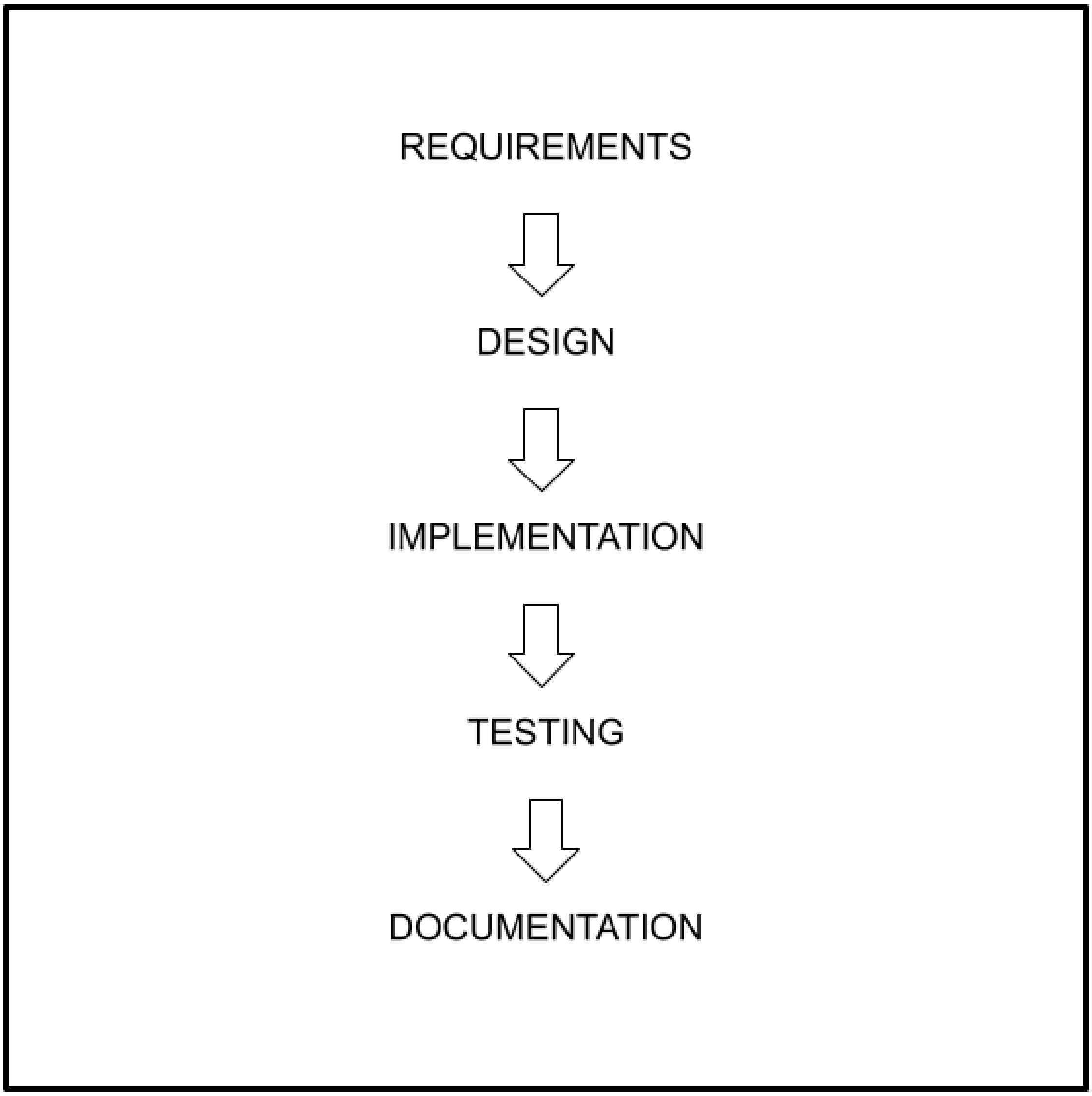
Effective inventory management is critical for retail operations. Traditional methods relying on manual tracking or real-time API synchronization with third-party platforms create operational friction. Web-based inventory systems with intuitive admin interfaces enable faster updates and reduce human error (Goel & Soni, 2016). The literature supports simplified stock status indicators (Available, Limited, Sold Out) as effective for customer-facing interfaces, balancing information richness with usability.

Sales analytics dashboards provide management with actionable insights for decision-making. Research shows that data-driven performance tracking improves operational efficiency and staff motivation (Davenport & Harris, 2007). Dashboard design principles emphasize clarity, real-time data updates, and task-relevant metrics rather than overwhelming users with information (Few, 2006).

Business intelligence and analytics systems support decision-making through data analysis and reporting (Turban, Sharda, & Delen, 2015). Integration of sales data, staff performance metrics, and operational information enables comprehensive business analysis and strategic planning.

7.0 METHODOLOGY

This project follows a Waterfall methodology with incremental demonstrations. The development lifecycle is divided into distinct phases: requirements, design, implementation, testing, and documentation. Each phase completes before the next begins, with three interim demonstrations (Week 6, 9, 12) to validate progress and gather feedback.



Waterfall Diagram 7.1

Phase 1 is Project Proposal and Planning (Week 1 – Week 4). The project scope, objectives, and system requirements are defined. Client consultations are conducted to confirm staff performance metrics and analytics requirements. The project plan and system design are prepared, while the development environment is configured using Visual Studio Code, Git, and GitHub. The Supabase project is initialized, including database configuration and authentication setup. The project proposal is presented in Week 4 (10%).

Phase 2 is System Design and Configuration (Week 5 – Week 6). The Supabase PostgreSQL database tables and relationships for products, stock, staff, sales, and administrator accounts are designed and configured. Low-fidelity mock-ups of the customer interface and admin dashboard are created. The user interface is designed using HTML5, CSS3, Bootstrap, and JavaScript. The system architecture and data flow between the frontend and Supabase backend are planned, and frontend development for customer-facing features begins. Demonstration 1 is conducted in Week 6, showcasing the product catalogue, stock status, and sizing reference tool (10%).

Phase 3 is Core Feature Development (Week 7 – Week 9). The admin dashboard is developed to support product management, stock updates, sales tracking, staff performance monitoring, and KPI reporting. Redirect links to Shopee and TikTok Shop are implemented. An AI-powered sales assistant chatbot is integrated into the admin dashboard using an AI provider API, enabling administrators to query sales-related topics and receive intelligent, context-aware responses. The frontend is connected to the Supabase backend via the Supabase JavaScript SDK, and data integration is tested to ensure all system components function correctly. Demonstration 2 is conducted in Week 9 to present the admin dashboard and additional features (10%).

Phase 4 is Testing and Refinement (Week 10 – Week 12). Manual functional testing is performed on all system features, including the customer interface, administrator operations, and database functionality. Integration testing is conducted between the frontend and Supabase backend. User Acceptance Testing (UAT) is carried out with client feedback where possible, followed by bug fixing and user interface improvements. Demonstration 3 (15%) and the Final Presentation (30%) are conducted in Week 12.

Phase 5 is Documentation (Week 13 – Week 15). The technical report is prepared to document the system architecture, database design, implementation, and testing process. The project logbook is completed by recording weekly progress, challenges, and solutions, while a user guide is prepared for administrator dashboard operations. The Technical Report (15%) and Logbook (10%) are submitted in Week 1.

8.0 REFERENCES

Chaffey, D., & Ellis-Chadwick, F. (2019). *Digital Marketing: Strategy, Implementation and Practice* (7th ed.). Pearson.

Davenport, T. H., & Harris, J. G. (2007). *Competing on Analytics: The New Science of Winning*. Harvard Business Review Press.

Few, S. (2006). *Information Dashboard Design: The Effective Visual Communication of Data*. O'Reilly Media.

Turban, E., Sharda, R., & Delen, D. (2015). *Business Intelligence and Analytics: Systems for Decision Support* (10th ed.). Pearson Education.

9.0 GANTT CHART

NO	Activity / Weeks	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1.1	Defining Project scope, objective and system requirements.															
1.2	Conducting client consultation															
1.3	Making the project proposal															
2.2	Supabase setup, database schema design, tables and relationships															
2.3	Frontend development (For Users)															
3.1	Admin dashboard development (product management, stock updates)															
3.2	Sales tracking, staff performance dashboard, KPI reporting, AI provider API chatbot integration															
3.3	Frontend-Supabase integration testing															
4.1	Manual functional testing (customer interface, admin operations)															
4.2	Integration testing, UAT with client, bug fixes, UI improvements															

4.3	Final polish															
5.1	Technical report writing															
5.2	Technical report finalization, logbook completion															

Discussion

Implementation

10.0 TOOLS, MATERIALS & COST

Item	**Quantity**	**Unit Cost (per year)**	**Total Cost (per year)**
Software & Tools	-	Free	RM0
Hosting	-	RM 180	RM180
Domain	-	RM 45	RM45
AI API Key	-	Free tier	RM0
Total:			RM225

11.0 CONCLUSION

In conclusion,the proposed **“Web-Based Integrated Operations Management System”** aims to improve the company's internal business operations by providing a centralized platform for managing sales records, inventory, employee performance, and Key Performance Indicators (KPIs). By replacing manual processes with a digital system, the project is expected to improve data accuracy, reduce administrative workload, and increase operational efficiency.

The system will also provide management with valuable insights through dashboards and reports, enabling better monitoring of business performance and supporting data-driven decision-making. These features will help the company identify sales trends, evaluate employee productivity, monitor inventory levels, and improve overall business management.

The significance of this project extends beyond meeting the current operational needs of the company. It demonstrates the practical application of information technology in solving real-world business problems while providing a scalable foundation for future enhancements such as advanced analytics, automated reporting, and integration with other business systems.

This project serves as an opportunity to apply the knowledge and technical skills acquired throughout the Diploma in Information Technology programme in developing a practical and beneficial solution for a real business environment.